

# ENGR 102 Summer Bridge

## Day 3 Student Quiz

## Raw Input, Type Evidence, Calculations, and F-Strings

20 points • approximately 20-25 minutes

| Name |  | UIN |  | Section |  |
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**Boolean-operator convention.** Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

**Question 1. Trace raw text, conversion, and formatted output.**

4 points

```
raw = "18.5"
value = float(raw)
adjusted = value + 1.5
print(type(raw).__name__, type(value).__name__)
print(f"{adjusted:.1f}")
```

Give the type and value at each named state, then write exact output.

**Question 2. Separate truthiness from an approval rule.**

4 points

```
approved_text = "no"
print(bool(approved_text))
print(approved_text == "yes")
```

Give both results and explain why a nonempty input string is not reliable approval evidence.

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**Question 3. Evaluate a multi-step Python math expression.**

4 points

```
side_a = 5
side_b = 12
diagonal = (side_a ** 2 + side_b ** 2) ** 0.5
print(f"{diagonal:.2f}")
```

Show the squared terms, the sum, the square-root step, and exact output.

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### Question 4. Program construction

8 points

Write one complete Python program that satisfies every requirement.

- Read distance and time as floating-point inputs.
- If time is not positive, print invalid and do not calculate speed.
- Otherwise compute  $\text{speed} = \text{distance} / \text{time}$ .
- Print Speed: followed by the result to exactly two decimal places.

[illegible]

Test evidence / expected result